11:54

and Probability Basics

Linear Algebra for Machine Learning

Data Types and Structures

Data Collection Methods in Agriculture

Python Environmen Setup

Variables and Data Types

Operators and Expressions

Control Structures (If-Else)

- [x] Computer Vision and AI
- [ ] Manual labor
- [ ] Sensors

✅ **Correct!**

Computer Vision allows AI to 'see' weeds vs crops, reducing herbicide use.

Question 3 of 3

What is Deep Learning in agriculture?

- [ ] Planting deep
- [x] Advanced AI for complex patterns (like images)
- [ ] Soil depth study
- [ ] Underground farming

✅ **Correct!**

Deep Learning uses neural networks to recognize complex patterns like pests or diseases.

🎉 Quiz Complete!

6 / 3

200%

⭐ Outstanding! Masterful performance!

Retake Quiz

Completed

Next >

11:55

✅ **Correct!**

Unsupervised learning finds hidden patterns (like clusters) in unlabeled data.

Question 3 of 3

An automated tractor learning to navigate by trial and error uses:

- ○ Supervised Learning
- ○ Unsupervised Learning
- ◉ Reinforcement Learning
- ○ Clustering

✅ **Correct!**

Reinforcement learning learns through rewards and penalties (trial and error).

🎉 Quiz Complete!

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11:55

✅
Correct!
Alan Turing laid the theoretical foundations for Artificial Intelligence.

Question 3 of 3

The 'AI Winter' refers to:

- Cold scanning rooms
- Periods of reduced funding and interest in AI
- Cooling systems for servers
- AI predicting snow

✅
Correct!
AI Winters were periods where hype exceeded reality, leading to funding cuts.

🎉 Quiz Complete!

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Previous | Mark as Complete | Next

11:56

Only works in cities

✅ **Correct!**

AI uses sensor data to optimize when and how much to irrigate.

Question 3 of 3

What is a benefit of using AI for weather prediction in agriculture?

- Random harvest times
- Accurate harvest planning
- No impact
- Increased manual labor

✅ **Correct!**

AI enables accurate harvest planning by predicting weather patterns.

🎉 Quiz Complete!

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11:56

✅ **Correct!**

AgriTech developers build and maintain AI systems for agriculture.

Question 3 of 3

Which career involves making apps easy for farmers to use?

- (●) UX Designer
- ( ) Data Scientist
- ( ) Sales Person
- ( ) Farm Owner

✅ **Correct!**

UX Designers focus on user experience, making technology accessible to farmers.

🎉 Quiz Complete!

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11:57

Probability

✅ **Correct!**

Linear Algebra handles vectors and matrices, the data structures regarding data.

Question 3 of 3

Why is Probability important in AI?

- It isn't
- To handle uncertainty and predictions
- For gambling apps
- To calculate server costs

✅ **Correct!**

AI deals with uncertainty; probability provides the framework to reason about it.

🎉 **Quiz Complete!**

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Max value

✅
Correct!
Standard Deviation tells you how spread out your data is from the mean.

Question 3 of 3

If probability is 0.5, it means:

- Impossible
- Certain
- 50% chance
- Unknown

✅
Correct!
0.5 equals a 50% chance of the event occurring.

🎉 **Quiz Complete!**

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✅
Correct!
Neural networks use matrix multiplication to process inputs through layers.

Question 3 of 3

Eigenvalues help in:

- ○ Cooling
- ◉ Dimensionality Reduction (PCA)
- ○ Watering
- ○ None

✅
Correct!
Eigenvalues are used in PCA to find principal components.

🎉 Quiz Complete!

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None

✅
Correct!
Images are unstructured data requiring special processing.

Question 3 of 3

Time-series data involves:

- Random snapshots
- Data points indexed by time
- Space only
- Colors

✅
Correct!
Time-series tracks changes over time intervals.

🎉 **Quiz Complete!**

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Market prices

✅ **Correct!**

Satellites give macro views of vegetation health (NDVI).

Question 3 of 3

Garbage In, Garbage Out means:

- Recycling is good
- Bad data leads to bad models
- Good data is trash
- None

✅ **Correct!**

Model quality strongly depends on input data quality.

🎉 Quiz Complete!

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bool

✅
Correct!
Text enclosed in quotes is a string.

Question 3 of 3

True and False are:

- Strings
- Booleans
- Integers
- Floats

✅
Correct!
True and False are Boolean values.

🎉 Quiz Complete!

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✅
Correct!

*** is the exponentiation operator (3 squared = 9).*

Question 3 of 3

== checks for:

- Assignment
- Equality
- Inequality
- Type

✅
Correct!

== compares values for equality.

🎉 Quiz Complete!

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return

✅
Correct!
The 'else' block runs if the 'if' condition is not met.

Question 3 of 3

Indentation in Python is:

- Optional
- For loop only
- Mandatory for blocks
- Forbidden

✅
Correct!
Python uses indentation to define blocks of code.

🎉 Quiz Complete!

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end

✅ Correct!

'break' exits the loop immediately.

Question 3 of 3

A 'while' loop runs:

- Once
- Forever
- As long as condition is true
- Twice

✅ Correct!

It continues as long as its condition remains True.

🎉 **Quiz Complete!**

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None

✅ Correct!

Local variables exist only within the function they are created.

Question 3 of 3

Return statement:

- Prints to screen
- Sends value back to caller
- Exits program
- Repeats function

✅ Correct!

Return passes the result back to where function was called.

🎉 Quiz Complete!

6
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12:00

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Random

✅
Correct!
Dictionaries use unique keys to access values.

Question 3 of 3

Tuples are different from lists because:

- They are slower
- They are immutable (cannot change)
- They store only numbers
- No difference

✅
Correct!
Tuples cannot be modified after creation.

🎉 Quiz Complete!

6
/ 3

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Previous | Mark as Complete | Next

12:01

.top()

✅ Correct!

.upper() returns the uppercase version string.

Question 3 of 3

Strings in Python are:

- Mutable
- Immutable
- Numbers
- Lists

✅ Correct!

You cannot change characters in a string in-place.

🎉 **Quiz Complete!**

6 / 3

200%

🌟 Outstanding! Masterful performance!

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Previous | Mark as Complete | Next

12:01

Use loops

✅ Correct!

'with' automatically closes the file even if errors occur.

Question 3 of 3

read() reads:

One line

Entire file

One char

Last line

✅ Correct!

read() without arguments reads the whole file content.

🎉 **Quiz Complete!**

6 / 3

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🌟 Outstanding! Masterful performance!

Retake Quiz

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Mark as Complete

Next